

Overview

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 |
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Overview

A discrete-event simulation study: the engine, the calibration and the experiments, with every number carrying its confidence interval.

MODELS 5	THROUGHPUT 0.807 items_per_minute	CYCLE TIME MEAN 14.8 minutes	UTILISATION 0.566
COMPLETED 2905	VALIDATION validated	REEVALUATION keep	

Design of the study

Seed 20260101
Days 45
Replications 8
Horizon 4000
Warmup 400
Models Any["mmc", "transfer_line", "machine_shop", "inventory", "call_center"]
Objective cycle_time_mean
Online policy offline_first

The models

Catalogue

Model	Title	Entity	Resource	Answers
mmc	ServicePool	customer	server	ServicePoolWithIdenticalServers
transfer_line	TransferLine	part	station	SerialLineWithBuffersAndFailures
machine_shop	JobShop	job	machine	JobShopWithRoutingsReworkScrapAndBreakdowns
inventory	InventoryPosition	order	stock	ReorderPointPolicyWithLeadTimeAndBackorders
call_center	ContactCenter	call	agent	AgentPoolWithPatienceAndPriority

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
availability	0.915	0.013	ratio	1.395
wait_p95	45.5	7.774	minutes	17.1
max_queue_length	3.522	0.968	count	27.5
queue_length_mean	9.583	1.474	count	15.4

Metric	Mean	Half width	Unit	Rel hw pct
cycle_time_mean	14.8	1.575	minutes	10.7
setup	2	0	minutes	0
scrapped	141.5	13.9	count	9.851
throughput	0.807	0.019	items_per_minute	2.398
events	27050	748.3	count	2.766
span	3600	0.122	count	0.00338
reworked	427.8	17.3	count	4.053
wait_mean	14.8	1.575	minutes	10.7
completed	2905	69.7	count	2.398
wip_mean	11.9	1.522	count	12.8
utilisation	0.566	0.013	ratio	2.219
tardiness_mean	9.855	1.548	minutes	15.7

DiscreteSim.jl, section overview, seed 20260101